AI-Powered Agriculture Crop Advisory Assistant

Evaluation Report — DeepEval / RAGAS / Custom Benchmarks

1. Information Retrieval Metrics (RAG Engine)

Metric	Score	Target	Status
Precision@5	0.83	> 0.80	PASS
Recall@5	0.78	> 0.75	PASS
Mean Reciprocal Rank (MRR)	0.925	> 0.90	PASS
Hit Rate@5	0.95	> 0.95	PASS
Avg Retrieval Latency	1.8 ms	< 5 ms	PASS

2. ML Model Performance

Metric	Score	Target	Status
Yield R^2 (Test Set)	0.9876	> 0.98	PASS
Yield MAE	0.31 t/ha	< 0.45	PASS
Yield RMSE	0.44 t/ha	< 0.60	PASS
Suitability Accuracy	94.7%	> 90%	PASS
Suitability F1	0.943	> 0.90	PASS
Inference Latency	< 5 ms	< 10 ms	PASS

3. Guardrail Evaluation

Test Category	Pass Rate	Target	Status
Injection Detection	100%	100%	PASS
Domain Scope Filter	98.6%	> 98%	PASS
Agronomic Range Check	100%	100%	PASS
Disclaimer Present	100%	100%	PASS

4. LLM-as-a-Judge Results

Dimension	Score	Target	Status
Relevance	0.96	> 0.90	PASS
Groundedness	0.98	> 0.95	PASS

Completeness	1.00	= 1.00	PASS
Safety	1.00	= 1.00	PASS

5. DeepEval RAG Metrics (Online Mode)

- Faithfulness: 0.97 — All generated claims traceable to retrieved cases
- Answer Relevancy: 0.95 — Advisory directly addresses query intent
- Contextual Relevancy: 0.89 — Retrieved context signal-to-noise ratio
- Contextual Precision: 0.91 — Top cases contain most informative content
- Contextual Recall: 0.93 — Retrieved context covers all necessary evidence

Note: DeepEval metrics requiring OpenAI API display as NOT AVAILABLE (Offline Mode) when running without API key. All deterministic metrics execute fully offline.